

Differential Calculus — Chain Rule

Chapter 2: Power Rule on Quadratic Inner Function | Page 33

PROBLEM STATEMENT

Power Rule & Quadratic Composite

Find the derivative with respect to x of the function:

$$y = (3x^2 + 4)^5$$

STEP 1 প্রদত্ত দ্বিঘাত ঘাত ফাংশনটি উল্লেখ করি (Given Composite Function)

$$y = (3x^2 + 4)^5$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(3x^2 + 4)^5]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\frac{dy}{dx} = 5(3x^2 + 4)^{5-1} \cdot \frac{d}{dx}(3x^2 + 4) = 5(3x^2 + 4)^4 \cdot \frac{d}{dx}(3x^2 + 4)$$

STEP 4 বন্ধনীভুক্ত অভ্যন্তরীণ অংশের অন্তরক নির্ণয় করি (Differentiating Inner Quadratic Expression)

$$\frac{d}{dx}(3x^2 + 4) = 3 \frac{d}{dx}(x^2) + \frac{d}{dx}(4) = 3(2x) + 0 = 6x$$

STEP 5 মান বসিয়ে ধ্রুবক ও চলক গুণ করে চূড়ান্ত অন্তরজ পাই (Multiplying Terms)

$$\frac{dy}{dx} = 5(3x^2 + 4)^4 \times 6x = 30x(3x^2 + 4)^4$$

FINAL ANSWER
$$\frac{dy}{dx} = 30x(3x^2 + 4)^4$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 02

🔴 BOX 1: FORMULATION & GENERALIZED POWER RULE

- **Core Formula:** $\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$.
- **Quadratic Composite Form:** For $(ax^2 + c)^n$, the derivative is:

$$\frac{d}{dx}(ax^2 + c)^n = n(ax^2 + c)^{n-1} \cdot (2ax) = 2anx(ax^2 + c)^{n-1}$$

- Here, $a = 3, c = 4, n = 5 \implies \frac{dy}{dx} = 2(3)(5)x(3x^2 + 4)^4 = 30x(3x^2 + 4)^4$.

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Leaving out the variable x in the inner derivative:** Writing $\frac{d}{dx}(3x^2 + 4) = 6$ instead of $6x$. This misses the power reduction of x^2 .
- **Multiplying inside the bracket prematurely:** Never write $30x(3x^2 + 4)^4$ as $(90x^3 + 120x)^4$; the exponent 4 prevents distribution of the external factor $30x$.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Domain of Definition:** The polynomial function is defined on $\mathbb{R} = (-\infty, \infty)$.
- **Differentiability:** Infinitely differentiable everywhere ($y \in C^\infty$).
- **Extrema & Stationary Points:** Setting $\frac{dy}{dx} = 0 \implies 30x(3x^2 + 4)^4 = 0$. Since $3x^2 + 4 \geq 4 > 0$ for all real x , the only real critical point is $x = 0$. At $x = 0, y = 4^5 = 1024$ (global minimum).

💡 BOX 4: LEIBNIZ SUBSTITUTION METHOD (ALTERNATIVE VERIFICATION)

- Let $u = 3x^2 + 4$, so $y = u^5$.
- $\frac{dy}{du} = 5u^4$.
- $\frac{du}{dx} = \frac{d}{dx}(3x^2 + 4) = 6x$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 5u^4 \cdot (6x) = 30xu^4 = 30x(3x^2 + 4)^4$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Stating the chain rule formula $\frac{d}{dx}(u^n) = nu^{n-1}u'$.
- **1.0 Mark:** Correct differentiation of inner quadratic function $6x$.
- **1.0 Mark:** Clean combination and final answer presentation $30x(3x^2 + 4)^4$.